

DATA ANALYTICS PROJECT

Excel · MySQL · Tableau · Power BI · Power Query

AdventureWorks Cycles

Sales Performance Analysis

Final Project Report

AdventureWorks Cycles | Multinational Sales Analysis | 2010–2014

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Chapter 1

Project Overview

1.1 About Adventure Works Cycles

Adventure Works Cycles is a large, multinational manufacturing company that designs and sells metal and composite bicycles across North American, European, and Asian commercial markets. Its base of operations is in Bothell, Washington, supported by 290 employees and regional sales teams worldwide.

In the 2000s, the company acquired a manufacturing plant in Mexico that produces critical subcomponents, which are shipped to Bothell for final assembly. By 2001, this plant had become the sole manufacturer and distributor of the touring bicycle product group.

Coming off a successful fiscal year, Adventure Works Cycles aims to broaden its market share by targeting its best customers, extending product availability through an external website, and reducing the cost of sales through lower production costs.

1.2 Headline Metrics

29.36M	12.08M	60.40K	41.15%
Total Sales	Total Profit	Total Orders	Profit Margin

1.3 Project Roadmap

The project followed four phases, forming one end-to-end analytics workflow:

Phase 1: Excel – Data cleaning, XLOOKUP/VLOOKUP, date fields, Sales & Production Cost formulas, pivot tables and charts.

Phase 2: MySQL Workbench – Importing Excel sheets, building tables with primary/foreign keys, joins, views, and SQL queries.

Phase 3: Tableau – Data relationships, calculated fields, KPI cards, filters, and an interactive dashboard.

Phase 4: Power BI – Power Query transformations, star-schema data modeling, DAX measures, and interactive reports.

Chapter 2

Dataset Overview and Data Model

2.1 Fact and Dimension Tables

The analysis is built on a star-schema data model consisting of one central fact table and six supporting dimension tables.

Table	Purpose
FactInternetSales	Main sales transactions
Fact_Internet_Sales_New	Additional sales transactions (unioned with the main fact table)
DimCustomer	Customer details
DimProduct	Product details
DimDate	Date table
DimSalesTerritory	Regional / territory details
DimProductCategory	Product category
DimProductSubCategory	Product sub-category

2.2 Star-Schema Design

The model links a single unified `Fact_Sales` table to six dimension tables (`DimCustomer`, `DimProduct`, `DimDate`, `DimSalesTerritory`, `DimProductCategory`, and `DimProductSubCategory`), enabling efficient slicing of sales data by customer, product, time, and geography.

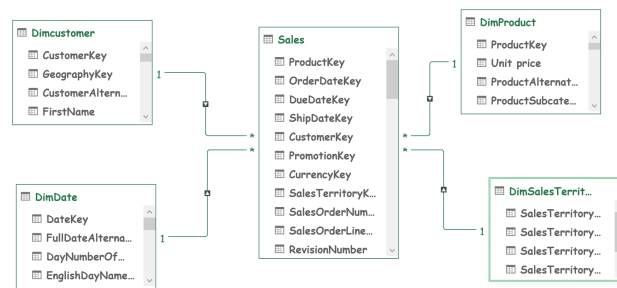


Figure 2.1: AdventureWorks star-schema data model – 1 fact table and 6 dimension tables.

Chapter 3

Data Preparation in Excel

The first phase of the project focused on cleaning and preparing the raw AdventureWorks data in Excel before it was modeled and visualized.

3.1 Key Preparation Steps

1. **Union Sales Data** – Combined `FactInternetSales` and `Fact_Internet_Sales_New` into a single unified `Sales` sheet.
2. **Lookups** – Used `XLOOKUP` / `VLOOKUP` to bring in Product Name, Customer Full Name, and Unit Price from dimension tables.
3. **Date Fields** – Derived Year, Month, Quarter, Year-Month, Weekday, and Financial Month/Quarter from `OrderDateKey`.
4. **Sales Amount** – Calculated as $UnitPrice \times OrderQuantity \times (1 - UnitDiscount)$.
5. **Production Cost** – Calculated as $UnitCost \times OrderQuantity$.
6. **Profit** – Calculated as $SalesAmount - ProductionCost$.

3.2 Project Questionnaire – Task Breakdown

The team's workflow was organized into three broad task groups:

Data Prep & Calculations	Visual Analysis	KPIs & Dashboard
Union the two sales tables	Pivot table: month-wise sales, filtered by Year	Additional KPIs by Product, Customer & Region
Lookup Product Name, Customer Name & Unit Price	Bar chart – year-wise sales	One or more consolidated dashboards
Build Year / Month / Quarter / Financial-period fields	Line chart – month-wise sales	
Calculate Sales Amount, Production Cost & Profit	Pie chart – quarter-wise sales	
	Combo chart – Sales Amount vs. Production Cost	

Chapter 4

SQL Analysis in MySQL Workbench

The second phase rebuilt the Excel workflow in SQL, using MySQL Workbench to create a reusable, queryable data foundation.

4.1 Data Foundation

A unioned view was created to serve as a single source of truth for all subsequent queries:

```
1 CREATE OR REPLACE VIEW v_UnionedSales AS
2   SELECT * FROM factinternetsales
3   UNION ALL
4   SELECT * FROM factinternetsalesnew;
```

Listing 4.1: Creating the unioned sales view

4.2 Analysis Categories

- **Data Foundation** – Union FactInternetSales + FactInternetSalesNew into v_UnionedSales; lookup ProductName from DimProduct; lookup CustomerFullName & Unit Price.
- **Calculated Fields** – Date parts from OrderDateKey (Year, Month, Quarter, Weekday, Financial Month/Quarter); Sales Amount = Price $\times$ Qty $\times$ (1 – Discount); Production Cost & Profit.
- **Chart-Ready Queries** – Monthly pivot filtered by Year; Year-wise & month-wise sales; Quarter-wise sales; Combo query of Sales Amount vs. Production Cost.
- **KPI Queries** – Top 10 Products by Sales; Top 10 Customers by Sales; Sales Performance by Region.

4.3 Advanced SQL Analytics

The team went beyond basic aggregation and built window-function and ranking queries to extract deeper insights.

Analysis	Description	Key SQL Construct
Year-over-Year Growth	Compares each year's sales to the prior year and computes YoY growth %	LAG() OVER (ORDER BY SalesYear)
Monthly Running Total	Cumulative sales total built month by month across the full order history	SUM(...) OVER (ROWS UNBOUNDED PRECEDING)
Best Product Category	Joins Product → SubCategory → Category and ranks total sales to find the top category	GROUP BY Category ORDER BY Sales DESC
Highest Profit Products	Top 10 products ranked by Sales Amount minus Production Cost	ORDER BY TotalProfit DESC LIMIT 10
Customer Ranking	Every customer ranked by total spend, with ties handled correctly	DENSE_RANK() OVER (ORDER BY Sales DESC)
Reusable View Layer	All 17 queries run against a single v_UnionedSales view – one source of truth for every chart	CREATE OR REPLACE VIEW v_UnionedSales

Chapter 5

Dashboards and Visual Analysis

Three interactive dashboards were built across Tableau, Power BI, and Excel, each offering a complementary view of the sales data.

5.1 Tableau Dashboard – Sales Overview

The Tableau dashboard focuses on year, quarter, and monthly sales trends together with top-performing products.

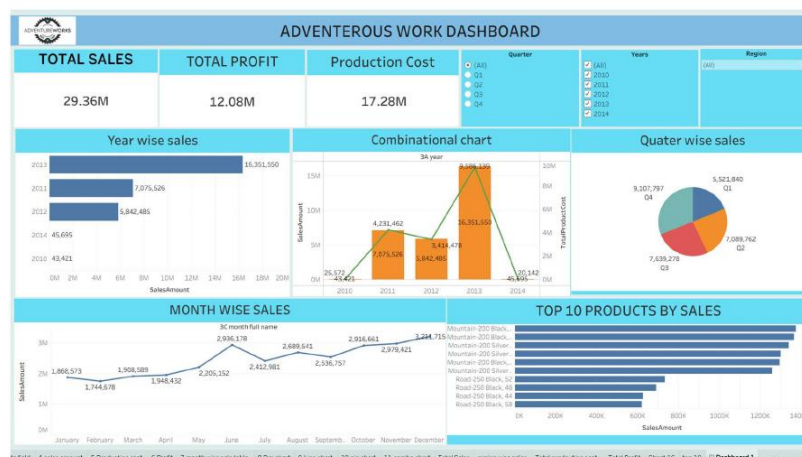


Figure 5.1: Tableau dashboard – sales overview with year, quarter, and monthly trends.

Visualizations used:

- **KPI Cards** – Total Sales, Total Profit & Production Cost at a glance
- **Bar + Line Combo** – Year-wise sales plotted against production cost
- **Pie Chart** – Quarter-wise sales distribution across Q1–Q4
- **Line Chart** – Month-wise sales trend across the year
- **Bar Chart** – Top 10 products ranked by sales amount
- **Slicers** – Filter the view by Quarter, Year & Region

5.2 Power BI Dashboard – Regional & Customer Insights

The Power BI dashboard emphasizes regional and customer-level insights with interactive slicers.

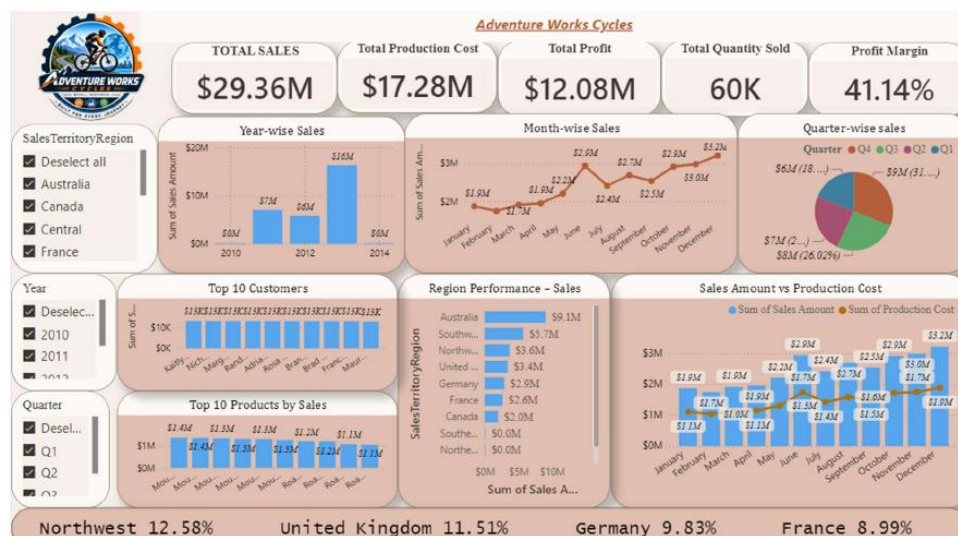


Figure 5.2: Power BI dashboard – regional and customer insights.

Visualizations used:

- **KPI Strip** – Sales, Cost, Profit, Quantity & Profit Margin
- **Bar Chart** – Year-wise sales split by sales territory
- **Line + Pie Charts** – Month-wise trend and quarter-wise sales share
- **Bar Charts** – Top 10 customers and top 10 products by sales
- **Combo Chart** – Sales amount vs. production cost by month
- **Slicers** – Region, Year & Quarter interactive filters

5.3 Excel Dashboard – Sales Performance Summary

The Excel dashboard provides a summary view of sales performance by year, quarter, and product.

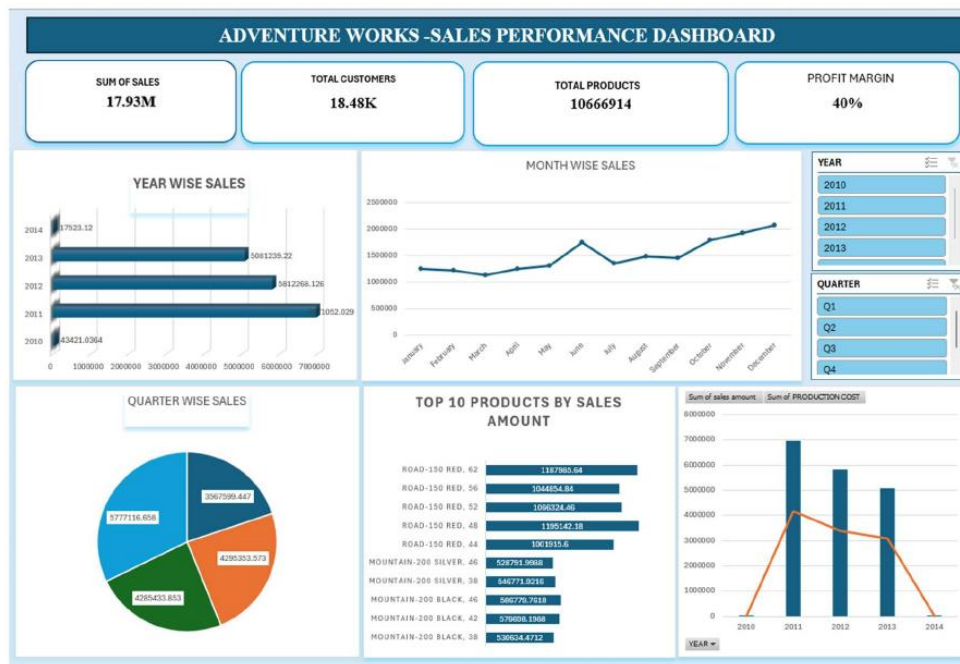


Figure 5.3: Excel dashboard – sales performance summary.

Visualizations used:

- **KPI Cards** – Sum of Sales, Customers, Products & Profit Margin
- **Bar Chart** – Year-wise sales trend, 2010–2014
- **Line Chart** – Month-wise sales pattern across the year
- **Pie Chart** – Quarter-wise sales contribution
- **Bar Chart** – Top 10 products by sales amount
- **Combo Chart** – Sales vs. production cost with Year & Quarter filters

Chapter 6

Key Insights

6.1 Headline Metrics (2010–2014)

29.36M	12.08M	60.40K	41.15%	\$486.09
Total Sales	Total Profit	Total Orders	Profit Margin	Avg Order Value

6.2 Top Findings

1. **Peak Year – 2013:** 2013 was the highest sales year at \$16M – 2.7× higher than 2012, with significant growth visible from 2011 onward.
2. **Top Region – Southwest:** Southwest USA led all regions in sales; Australia was the top international market, generating \$9.06M in sales.
3. **Top Product:** The Mountain-200 Black (42) was the best seller at \$1.37M – mountain bikes dominate the top 10 products list.
4. **Top Customer:** Jordan Turner leads with \$16.00K in purchases; the top 10 customers are closely ranked between \$13K and \$16K.
5. **Q4 Peaks Sales:** Q4 is consistently the strongest quarter – the holiday-season effect is visible across all years.
6. **Sales Exceed Cost:** Production cost stays well below the sales amount across all months, reflecting a healthy and stable profit margin of 41.15%.

Chapter 7

Tools and Technology Stack

The project drew on a complete, industry-standard analytics toolkit spanning data preparation, database analysis, and interactive visualization.

Tool		Role in the Project
Excel	Work-bench	Data cleaning, lookups, pivot tables, charts & dashboards
MySQL		Table design, joins, views & SQL queries
Tableau		Data relationships, calculated fields & interactive dashboards
Power BI		Power Query, data modeling, DAX measures & reports

Chapter 8

Conclusion

This project delivered a complete, end-to-end analytics workflow for AdventureWorks Cycles' multinational sales data, spanning 2010–2014. Starting from raw transactional data, the team progressed through data cleaning and calculation in Excel, relational modeling and advanced querying in MySQL Workbench, and interactive dashboard development in Tableau, Power BI, and Excel.

The resulting analysis identified 2013 as the strongest sales year, the Southwest USA and Australia as leading markets, mountain bikes as the top-performing product category, and a consistent Q4 seasonal peak driven by holiday demand. With total sales of \$29.36M, total profit of \$12.08M, and a healthy profit margin of 41.15%, AdventureWorks Cycles demonstrates a stable and profitable multinational sales operation, with clear opportunities to further grow market share among its best customers and regions.

Team – Group 4

Excel · SQL · Tableau · Power BI